CS 255 Business Requirements Document Template

TRISTAN PONTE

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of this project is to develop a web based application for DriverPass to manage student training and lesson scheduling.
- The system is being designed to fill a gap in the market by providing students with the tools they need to pass DMV driving tests through online classes and on the road training and help mitigate past failures.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- The main problem is a significant 65% failure rate for DMV driving tests which indicates a need for structured and accessible driver education.
- The solution is a centralized platform where students can access practice materials, purchase training tiers, and coordinate live driving sessions.
- Core components are going to be a cloud hosted web server, a secure customer database, an administrative dashboard for tracking, and a notification link with the DMV.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- Digital Enrollment: Enable 100% online registration and training package selection (6, 8, or 12 hour tiers)
- Resource Optimization: Automated tracking of 10 cars and their assigned drivers to prevent scheduling conflicts.
- Accountability: Maintain a verifiable audit trail of every record modification (reservations, cancellations, etc...)
- Compliance: Establish a live notification feed from the DMV to ensure all practice tests remain current with state policies..

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system must run as a web based application with responsive loading times for both desktop and mobile users so that it is seamless for all end users.
- Database queries for availability must return results in under 5 seconds to prevent user dropoff.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- Must be a cloud hosted solution to minimize client side maintenance (backups and security).
- Compatibility required for modern web browsers (Chrome, Safari, Edge) and mobile OS (iOS, Android)

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system will need to be able to distinguish between roles (Owner, IT, Secretary, Student) to enforce appropriate access rights.
- Financial data (credit card numbers, exp dates) must be captured with 100% precision to prevent transaction errors.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- Admins must be able to "disable" existing training packages without requiring code changes.
- The system should be able to scale to allow for future additions of new training modules or vehicles.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- Secure data exchange between client and server using SSL/TLS encryption.
- Role-based access control (RBAC) to allow the IT officer to block accounts or reset passwords.
- "Brute-force" protection that locks accounts after five failed login attempts.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall allow students to create and manage secure online accounts.
- The system shall process lesson reservations for two hour sessions.
- The system shall allow users to cancel or modify appointments online.
- The system shall identify and match a specific car, driver, and student for every session.
- The system shall display test progress with statuses like "not taken," "in progress," "failed," or "passed".
- The system shall capture student driver comments for review after each session.
- The system shall generate Excel compatible reports for administrative data analysis.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

What are the needs of the interface?

- The interface needs to be a web based, cloud hosted platform that is accessible across various devices, ensuring a consistent user experience whether the user is in the office or on the road.
- It must feature a centralized dashboard layout as sketched by Liam, which provides a high level view of student progress, lesson notes, and personal profiles in one screen.
- The system needs specific input forms for registrations and contacting support designed to minimize manual data entry errors.
- A notification system is required to alert users and admins of real time updates from the DMV regarding new rules or test questions.

Who are the different users for this interface?

- The Owner (Liam): Will need high level reporting views and the ability to toggle training package availability.
- IT Officer (Ian): Requires an administrative backend for account management, password resets, and security monitoring.
- Secretary: Needs a focused interface for answering calls, registering new students, and managing the lesson schedule.
- Students (Customers): Will require a self service portal to book lessons, take practice tests, and track their pass/fail status.
- Drivers (Trainers): Need a simplified interface to record comments on student performance and log lesson hours.

What will each user need to be able to do through the interface?

- Students will use the interface to view their online test progress, specifically tracking scores and the status of tests (like passed, failed, or in-progress...)
- Drivers will use the "Driver Notes" module to input comments and verify lesson start and end times for each session.
- Internal Staff will use data entry forms to capture student names, addresses, credit card info, and specific pickup / drop off locations.
- IT Support will use the interface to manage the automatic password reset flow for customers who have been locked out.

How will the user interact with the interface (mobile, browser, etc.)?

- The primary interaction will be through standard web browsers on both desktop and laptop computers for office staff and students taking online classes.
- The interface must be mobile responsive, allowing Liam to access reports and drivers to input notes while on the road via smartphones or tablets.
- Liam specifically requested the ability to download reports from the interface to work on them locally using software like Excel.

Assumptions

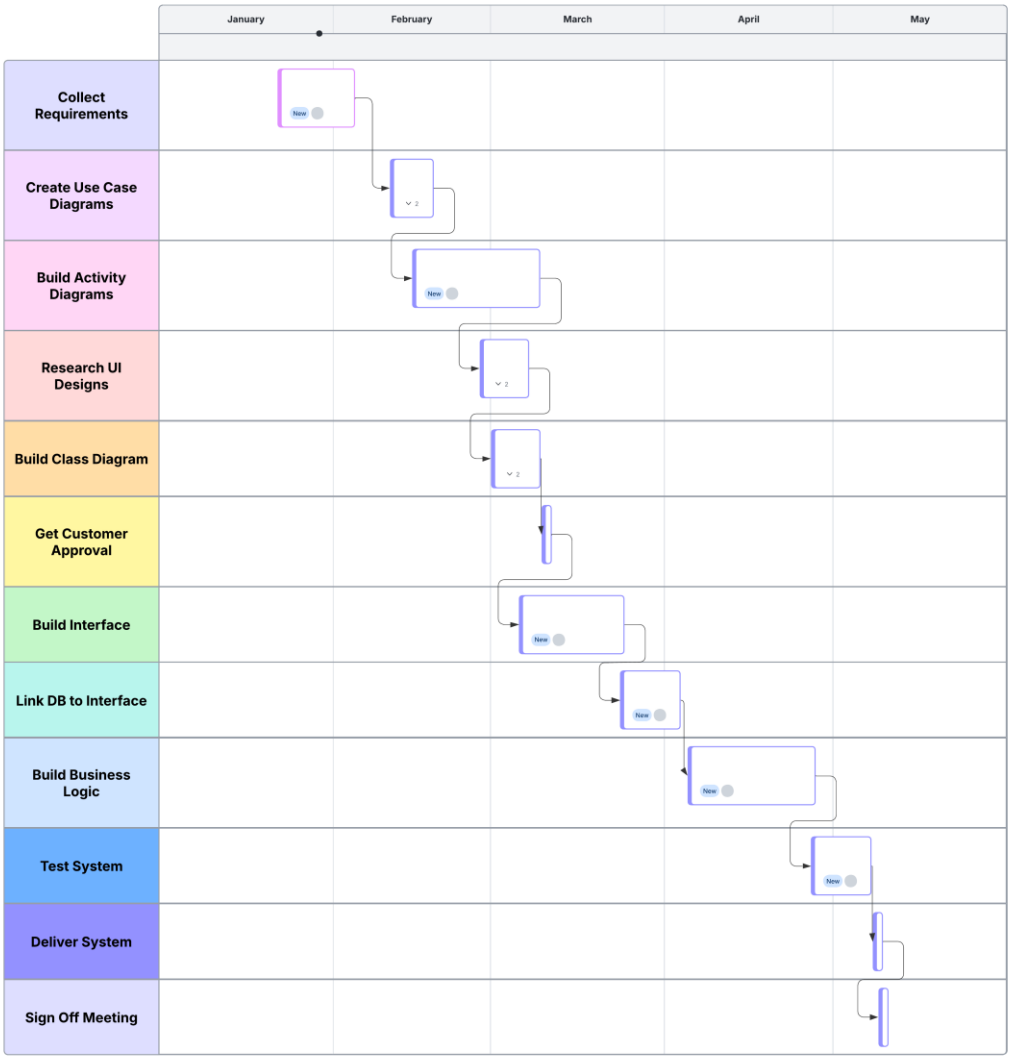
What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- Users have consistent internet access for scheduling and modifying data to prevent redundancy.
- The DMV provides a standard API or feed for rule updates and practice questions.
- All 10 vehicles are equipped with a driver/trainer at all times during operating hours.

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- Modification Restriction: Users cannot modify or update data while offline.
- Customization: Non developers cannot add entirely new modules; they can only disable existing ones.
- Vehicle Cap: The current system design is optimized for 10 cars and scaling beyond this may require backend resource adjustments which shouldnt be difficult since this will be on the cloud which will make scaling much easier.

Gantt Chart

DriverPass Gantt Chart